



KARNATAKA SSLC PATTERN - MATHEMATICS PRACTICE QUESTION PAPER

S.S.L.C. / CLASS X - MATHEMATICS

English Medium | Subject Code: 81-E

Time: 3 Hours 15 Minutes	Max. Marks: 80
Full Syllabus	Total Questions: 38

General Instructions to the Candidate

1. This question paper consists of **38 questions**. Follow the instructions given against each section.
2. Figures in the right-hand margin indicate the maximum marks allotted.
3. The maximum time includes **15 minutes for reading** the question paper.
4. Answer all questions unless an **OR** choice is provided.
5. This is an original practice paper and is not an official KSEAB paper.

I. Four alternatives are given for each question. Choose the correct alternative. $8 \times 1 = 8$

1. The HCF of 96 and 144 is
(A) 12 (B) 24 (C) 48 (D) 72
2. The zeroes of $x^2 - 5x + 6$ are
(A) 1, 6 (B) 2, 3 (C) -2, -3 (D) 3, 6
3. The pair $2x+3y=7$ and $4x+6y=14$ has
(A) no solution (B) one solution (C) infinitely many solutions (D) two solutions
4. The 10th term of the AP 3, 7, 11, 15,... is
(A) 35 (B) 39 (C) 40 (D) 43
5. If two triangles are similar, the ratio of their areas is equal to
(A) ratio of corresponding sides (B) square of ratio of corresponding sides (C) cube of ratio of corresponding sides
(D) sum of corresponding sides
6. The distance between (0,0) and (6,8) is
(A) 8 (B) 10 (C) 12 (D) 14
7. The probability of getting an even number on a fair die is
(A) $1/6$ (B) $1/3$ (C) $1/2$ (D) $2/3$
8. The curved surface area of a cylinder of radius r and height h is
(A) $2\pi r^2$ (B) $\pi r^2 h$ (C) $2\pi r h$ (D) $\pi r h^2$

II. Answer the following questions. $8 \times 1 = 8$

9. Use Euclid's division algorithm to find the HCF of 135 and 225.
10. Find the value of k if one zero of $2x^2 + kx + 3$ is 1.
11. Solve $x + y = 9$ and $x - y = 3$.
12. Find the discriminant of $3x^2 - 5x + 2 = 0$.
13. Find the 15th term of the AP 5, 9, 13, 17,...
14. Find the coordinates of the midpoint of A(4,-2) and B(-6,8).
15. If $\sin \theta = 3/5$ for an acute angle θ , find $\cos \theta$.
16. Find the mean of 4, 7, 9, 10 and 15.

III. Answer the following questions. $8 \times 2 = 16$

17. Prove that $\sqrt{3}$ is irrational.
18. Solve by elimination: $2x+3y=13$ and $4x-3y=5$.
19. Find the zeroes of $2x^2 - 7x + 3$ and verify the relationship between zeroes and coefficients.
20. Solve $x^2 - 9x + 20 = 0$ by factorisation.
21. Find the sum of the first 20 terms of the AP 7, 10, 13,...
22. In $\triangle ABC$, $DE \parallel BC$ with D on AB and E on AC. If $AD:DB = 2:3$, find $AE:EC$.
23. Find the area of the triangle with vertices (2,3), (5,7) and (7,3).
24. A bag contains 5 red, 3 blue and 2 green balls. One ball is drawn at random. Find the probability that it is (i) blue or green, (ii) not red.

IV. Answer the following questions. $9 \times 3 = 27$

25. Solve by cross-multiplication: $3x+2y=12$ and $5x-y=9$.
26. The 7th term of an AP is 19 and the 13th term is 37. Find the first term and common difference.
27. A tangent is drawn from an external point P to a circle with centre O. If $OP=13$ cm and radius=5 cm, find the tangent length.
28. Prove the Basic Proportionality Theorem.
29. A tower is observed from a point 30 m away. The angle of elevation of its top is 60° . Find the height of the tower.
30. Find the area of a sector of radius 14 cm and angle 90° . Also find its arc length.
31. A solid consists of a cylinder of radius 3 cm and height 8 cm with a hemisphere of the same radius on top. Find its total volume.
32. The following marks are 12, 15, 18, 20, 22, 25, 28, 30. Find the mean and median.

33. Two coins are tossed together. Find the probability of getting (i) exactly one head and (ii) at least one head.

V. Answer the following questions. **$4 \times 4 = 16$**

34. Solve graphically or algebraically: $2x+y=7$ and $x-y=2$. Verify the solution.

35. A rectangular field has area 528 m^2 . Its length is 10 m more than its breadth. Form the quadratic equation and find its dimensions.

36. From a point on the ground, the angle of elevation of the top of a 20 m tower is 30° . Find the distance of the point from the tower. Also find the angle of elevation if the distance is reduced to $20/\sqrt{3}$ m.

37. A solid toy is made of a cone surmounted by a hemisphere. Both have radius 3.5 cm and the cone has height 12 cm. Find the total surface area and volume.

VI. Answer the following question. **$1 \times 5 = 5$**

38. A vessel is in the form of an inverted cone of height 12 cm and radius 6 cm. It is filled with water to a height of 8 cm. Find the volume of water. If the water is poured into a cylinder of radius 4 cm, find the height of water in the cylinder.

SYLLABUS COVERAGE & PAPER BLUEPRINT

The paper covers the Karnataka Class 10 Mathematics areas reflected in the SSLC syllabus: Real Numbers; Polynomials; Pair of Linear Equations in Two Variables; Quadratic Equations; Arithmetic Progressions; Triangles; Coordinate Geometry; Introduction to Trigonometry; Some Applications of Trigonometry; Circles; Areas Related to Circles; Surface Areas and Volumes; Statistics; Probability.

END OF QUESTION PAPER